

Cloud Physics, Fall 2022 Homework 2

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1. Supersaturation Development Equation

將飽和發展方程同乘 dt ，並對兩邊做積分，從雲底積到雲內，可得下式。

$$\frac{ds}{dt} = A_1 \cdot \frac{dz}{dt}, \quad A_1 = 5.5 \cdot 10^{-4} \text{m}^{-1}$$

$$ds = A_1 \cdot dz$$

$$\int ds = A_1 \cdot \int dz$$

$$s - s_0 = A_1(z - z_0)$$

將雲內高度與飽和度條件帶入方程，並使雲底飽和度為 100%，得雲底高度。

$$(105.5 - 100)\% = 5.5 \cdot 10^{-4}(300 - h_1) \Rightarrow h_1 = 200\text{m}$$

$$(102 - 100)\% = 5.5 \cdot 10^{-4}(500 - h_2) \Rightarrow h_2 \approx 463.64\text{m}$$

$$(106.9 - 100)\% = 5.5 \cdot 10^{-4}(800 - h_3) \Rightarrow h_3 \approx 674.5\text{m}$$

2. Bowen's Model

$$\frac{dR}{dz} = \frac{dR}{dt} \frac{dt}{dz} = \frac{\bar{E} \cdot L}{4\rho_w} \cdot u(R) \cdot \frac{1}{w - u(R)}$$

$$E = 1, L = 2.5\text{gm}, u(R) = aR^b \text{ where } a = 8500\text{s}, b = 1$$

	w [m/s]	r_0 [μm]		w [m/s]	r_0 [μm]
A	2	15	C	4.5	15
B	2	40	D	4.5	40

(1) Calculate the radius of the droplet(s) when it falls to the cloud base.

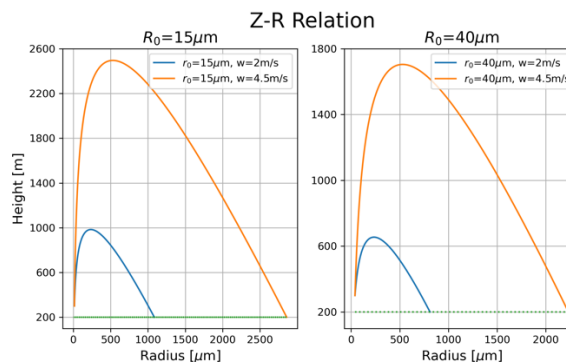
A : $r_0 = 15 \mu\text{m}$, $w = 2 \text{ m/s}$, the radius = 1084.8826553205597 μm

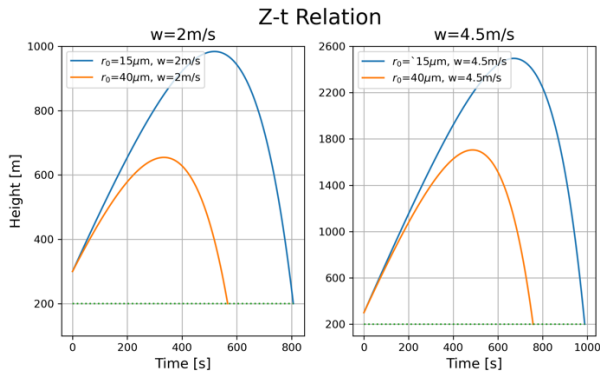
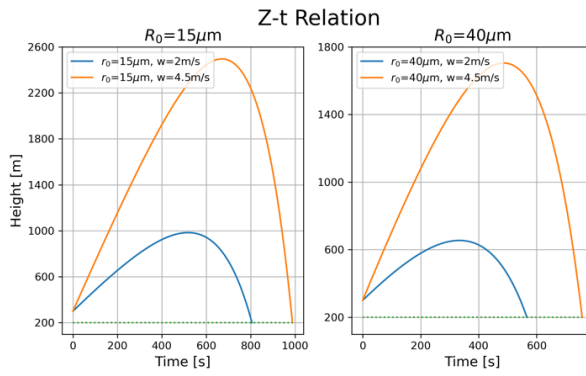
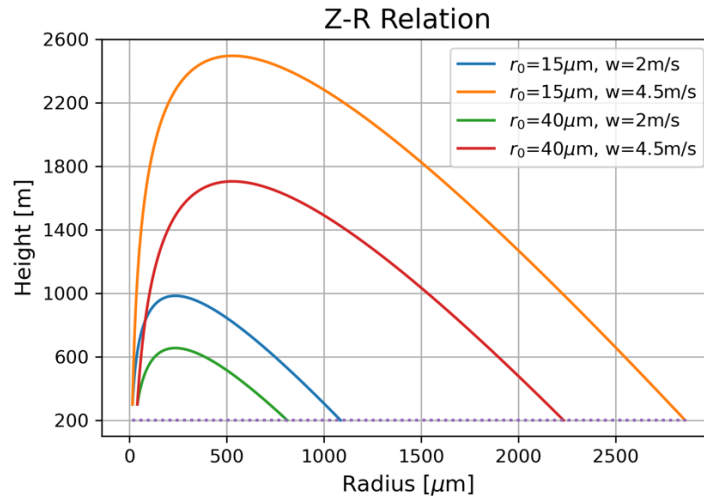
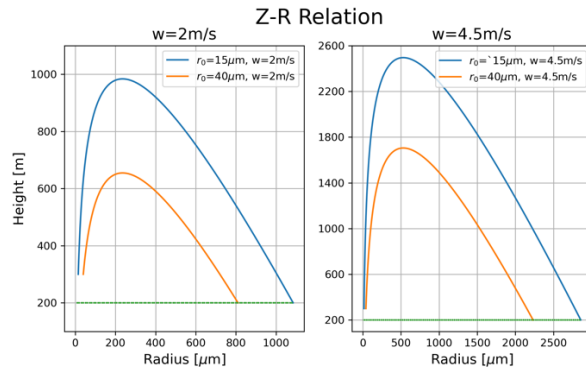
B : $r_0 = 40 \mu\text{m}$, $w = 2 \text{ m/s}$, the radius = 810.4048295956056 μm

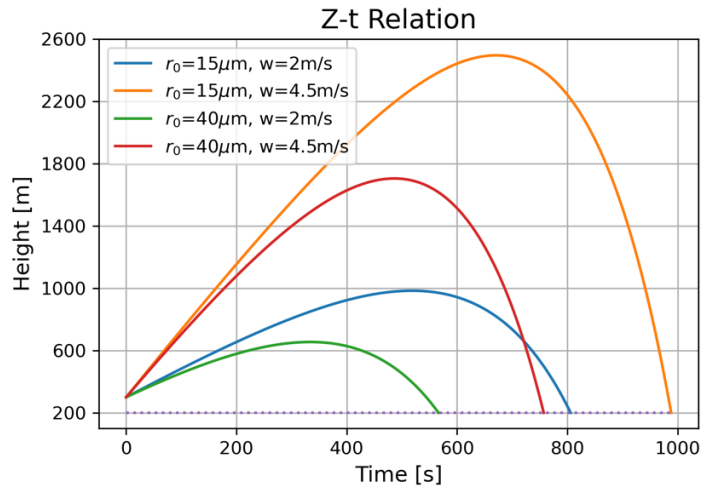
C : $r_0 = 15 \mu\text{m}$, $w = 4.5 \text{ m/s}$, the radius = 2856.648936645826 μm

D : $r_0 = 40 \mu\text{m}$, $w = 4.5 \text{ m/s}$, the radius = 2231.5269974119374 μm

(2) Plot the Z-t diagram and Z-R diagram







(3) Compares the results and discuss.

由 Z-R diagram 可以發現：相同初始半徑下，當雲內上升速度愈快時，雨胚會被抬升到雲內愈高的位置，同時能吸收更多雲內水氣，雨胚成長的半徑也愈大。相同上升速度下，初始半徑小的雨胚更有優勢成長成較大的雨胚。但綜觀來看，上升速度比起初始半徑更主導了雨胚成長的大小。

由 Z-t diagram 可以發現：相同初始半徑下，雲內上升速度愈快，雨胚被抬升位置也會更高，同時花費較多的時間。相同上升速度下，小雨胚比大雨胚花更多時間成長。

綜觀來看，雨胚成長花費時間由上升速度和初始半徑共同主導。

3. Continue from problem 2, for the four conditions, calculate the time for the droplet(s) to fall from the cloud base to the surface ($Z = 0$), and the radius of the droplet(s) at the surface.

Compare the results and discuss.

A : $r_0 = 15 \mu\text{m}$, $w = 2 \text{ m/s}$, time from cloud base to ground = 21.6899999999 s

B : $r_0 = 40 \mu\text{m}$, $w = 2 \text{ m/s}$, time from cloud base to ground = 29.0399999999 s

C : $r_0 = 15 \mu\text{m}$, $w = 4.5 \text{ m/s}$, time from cloud base to ground = 8.2399999999 s

D : $r_0 = 40 \mu\text{m}$, $w = 4.5 \text{ m/s}$, time from cloud base to ground = 10.5499999999 s

A : $r_0 = 15 \mu\text{m}$, $w = 2 \text{ m/s}$, the radius = 1084.8249740062279 μm

B : $r_0 = 40 \mu\text{m}$, $w = 2 \text{ m/s}$, the radius = 810.4047400106776 μm

C : $r_0 = 15 \mu\text{m}$, $w = 4.5 \text{ m/s}$, the radius = 2856.4971780211185 μm

D : $r_0 = 40 \mu\text{m}$, $w = 4.5 \text{ m/s}$, the radius = 2231.5269855924794 μm

形成雨滴後粒徑變大，從球體 rigid sphere 變成變形狀 deformed sphere，此時底面積大風阻多，但終端速度依然正比於半徑 $u_{\infty} \propto r^{1/2}$ ，可見大粒子終端速度大掉落花費時間少。將地面的雨滴半徑除上雲底時半徑可見雨滴再掉落過程的變化。

	$r(z=0)/r(z=200)$	Time [s]		$r(z=0)/r(z=200)$	Time [s]
A	99.9947%	21.69	C	99.9999%	8.24
B	99.9947%	29.04	D	99.9999%	10.55

Mason equ.中並沒有考慮碰著過程，且雨滴從雲底下落到地面的過程中，水密度並無在雲裡來的大，所以我們僅需考慮蒸發和凝結。離開雲底後，空氣的水氣壓小於水滴表面的水氣壓，因此雨滴會發生些微的蒸發，對照上表可以發現，掉落過程越久的雨滴，比起原在雲底時變得更小，蒸發更多。

